Varshith Sreeramdass

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Research Statement

My research focuses on agile human-robot collaboration: robot control informed by a partner's motion, toward robots that work alongside people. I study this on two platforms: a robotic wheelchair tennis partner and adaptive exoskeletons. My work combines hierarchical reinforcement learning, generative modeling, and sim-to-real transfer. I am seeking internships for summer 2027, with the long-term goal of an industry research career deploying collaborative robots.

Education

PhD in Robotics — *Atlanta, US*
Georgia Institute of Technology | GPA: 4.0/4.0 — *2024 – 2029**
- Advised by Prof. Matthew Gombolay and Prof. Aaron Young. — **Expected*
- Recipient of the IRIM Robotics PhD Fellowship.

MS in Computer Science — *Atlanta, US*
Georgia Institute of Technology | GPA: 4.0/4.0 — *2022 – 2024*
- Specialized in Computational Perception and Robotics.
- Coursework: DL for Robotics, Mobile Manipulation, ML with Graphs, Humans and ML.

B.Tech. in Computer Science and Engineering with Honors — *Mumbai, India*
Indian Institute of Technology Bombay (IITB) | GPA: 8.72/10.0 — *2015 – 2019*
- Thesis advised by Prof. Sunita Sarawagi and Prof. Soumen Chakrabarti.
- Coursework: Advanced ML, Foundations of Intelligent Learning Agents, Neuromorphic Engineering, Graphics.

Experience

Honda R&D, Frontier Robotics — *Tokyo, Japan*
Research Engineer — *2019 – 2022*
- Developed offline and residual RL for robust dexterous reorientation and can-opening.

Honda Research Institute — *Tokyo, Japan*
Summer Internship — *2018*
- Explored sequence-to-sequence models for sign language motion synthesis.

Publications

[P1] Generalized Behavior Learning from Diverse Demonstrations
V. Sreeramdass, R. R. Paleja, L. Chen, S. van Waveren, and M. Gombolay.
International Conference on Learning Representations (ICLR), 2025. [Paper] [Code] [Poster]
Out-of-Distribution Generalization Workshop at the Conference on Robot Learning (CoRL), 2023 **(Oral)**. [Paper]

Active Research

Agile and Robust Behavior Learning for a Wheelchair Tennis Robot Partner — *GaTech*
Prof. Matthew Gombolay — *Aug '24 – Present*
- Designed hierarchical RL policies for tennis, decoupling swing and ball placement [swing] [strokes] [juggling].
- Sustained juggling for 140 consecutive real-world ball strikes until a 2-min trial cap via zero-shot sim-to-real transfer from Isaac Lab, stabilized by a Gaussian-process model of the racket's spatially varying restitution [video].
- Evaluating preliminary human-robot teaming via collaborative juggling with data-driven proxy teammates.

Exoskeleton Foundation Models for Unifying Sensing, Joint States, and Control — *GaTech*
Prof. Aaron Young, Prof. Matthew Gombolay — *Jan '26 – Present*
- Designing a time-series diffusion model over onboard sensors, joint states, and biological and assistance torques.
- Using diffusion-based imputation to condition on any available subset of signals and generate the rest, enabling one model across devices with differing sensor suites, and inference-time steering of assistance.
- Distilling sensor→torque mappings into single-pass models, targeting real-time control on NVIDIA Jetsons.

Past Research

Task-Agnostic Exoskeleton Control from Simulated and Real Biomechanics Data *GaTech*

Prof. Aaron Young, Prof. Matthew Gombolay *Aug '24 – Dec '25*

- Developed task-agnostic control using transformer and diffusion models conditioned on long-horizon motion context.
- Used variational inference to disentangle task- and subject-specific factors toward personalized assistance.
- Scaled training using simulated sensors via unpaired sim-to-real translation (CycleGANs, Schrödinger bridges).

Generalized Behavior Learning from Diverse Demonstrations *GaTech*

Prof. Matthew Gombolay | MS Project [P1] *Jan '23 – Jun '24*

- Investigated inverse RL for discovering behaviors from diverse demonstrations that generalize to unseen latent factors.
- Developed latent regularization to emphasize diversity over demonstrated regions, resulting in task-relevant behaviors.
- Improved behavior discovery by $\sim$21% in latent factor errors across four simulated domains, including table tennis.

Dexterous Manipulation using Data-driven and Residual Reinforcement Learning *Honda R&D*

Prof. Takayuki Osa, Akinobu Hayashi, Tadaaki Hasegawa *Apr '20 – Jun '22*

- Investigated demonstration-based, offline, and residual RL for reorientation and manipulation of articulated objects.
- Achieved reorientation robust to $\pm$4cm and $\pm$30° initial-pose noise with offline RL and online finetuning [Press] [Video].
- Improved robustness of scripted policies for soda can-opening with residual RL; demonstrated at IROS '22 [Video].

Honors and Awards

Top Reviewer,	Conference on Neural Information Processing Systems (NeurIPS).	2026
IRIM Robotics PhD Fellowship,	Institute for Robotics and Intelligent Machines, Georgia Tech.	2024
TA of the Month,	CSE Department, IIT Bombay; selected among $\sim$100 TAs.	2018
All India Rank 204,	JEE Advanced, among 150,000 candidates shortlisted from JEE Main.	2015
KVPY Fellowship,	All India Rank 710; awarded by the Government of India.	2014

Skills

RL Algorithms Online: PPO, SAC, TD3; Unsupervised: DIAYN, CSD; Data-driven/Inverse: GAIL, AIRL, DAPG.
ML Models Diffusion, Transformers, Gaussian Processes, Autoencoders, CycleGANs, Schrödinger Bridges.
Robotics/ML Frameworks ROS(2), Isaac Lab, MuJoCo, PyBullet, Pinocchio, PyTorch, TensorFlow.
Miscellaneous Ray Tune, wandb, SHAP, pygmo, Git, pybind11, OpenGL.

Leadership and Service

Organization

- Co-organizer, RoboLetics 3.0 Workshop at CoRL 2026 (Austin, US).

Peer Review

- NeurIPS 2026 (Top Reviewer); CoRL 2025, 2026; IROS 2026; ISRR 2026; HRI 2025.

Teaching Assistantships

Interactive Robot Learning | GaTech *Fall '22, Spr '24*

- Designed and graded assignments and exams; held office hours for assignments and projects. Class of $\sim$50.

Introduction to Cognitive Science | GaTech *Fall '23*

- Graded assignments and projects. Class of $\sim$300.

Software Systems Lab | IITB *Fall '18*

- Co-led the TA team; delivered guest lectures, designed and graded assignments and exams. Class of $\sim$100.

Co-founder, Synergy From Diversity

Frontier Robotics | Honda R&D *Apr '21 – Apr '22*

- Organized language exchanges and cultural workshops for Japanese and international employees during COVID-19.

Academic Mentor

CSE Department Academic Mentorship Program | IITB *May '18 – May '19*

- Mentored sophomores and organized department-wide remedial classes to complement course curricula.
- Advised senior students with academic backlogs under the Academic Rehabilitation Program.

Miscellaneous

I speak English, Japanese (N3), Hindi, and Telugu. I enjoy cooking, hiking, watching movies, and collecting vinyl.